

Bat Face Recognition Using Deep Learning

Sagi Levi, 07-2025



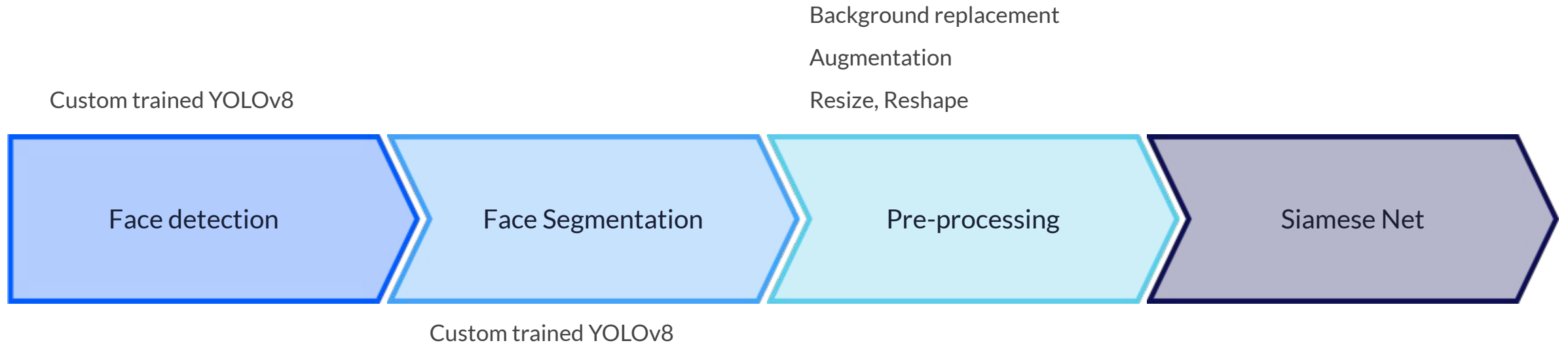
AGENDA

Introduction

This research uses a Siamese neural network to test whether Mauritanian bats have evolved distinct facial cues due to sleeping in light, unlike Rousettus bats that sleep in darkness.

1. Image Processing Pipeline
2. Experiments Overview (Mauritius)
3. Saliency Map Discovery
4. Confusion Matrix
5. Final Model Performance
6. Rousettus Comparison
7. ML Flow

Processing Pipeline



Network Architecture



Experiments

- Multi-Angle + Augmentation

- Using all cutouts from videos
- Augmentation for data enrichment (x60)

- Frontal Only

Using only frontal images (no augmentation)

- Green background

Replace the frontal images backgrounds with green background

- Random Background

Replace the frontal images backgrounds with green background



Experiments Results (Mauritius)

- Multi-Angle + Augmentation

High success rate, but maybe biased

- Frontal Only

High success rate, but saliency showed the model doesn't focus on the face

- Green background

Low success rate, focus on a green background as a preliminary condition

- Random Background

High success rate (~87%). Saliency shows use of face



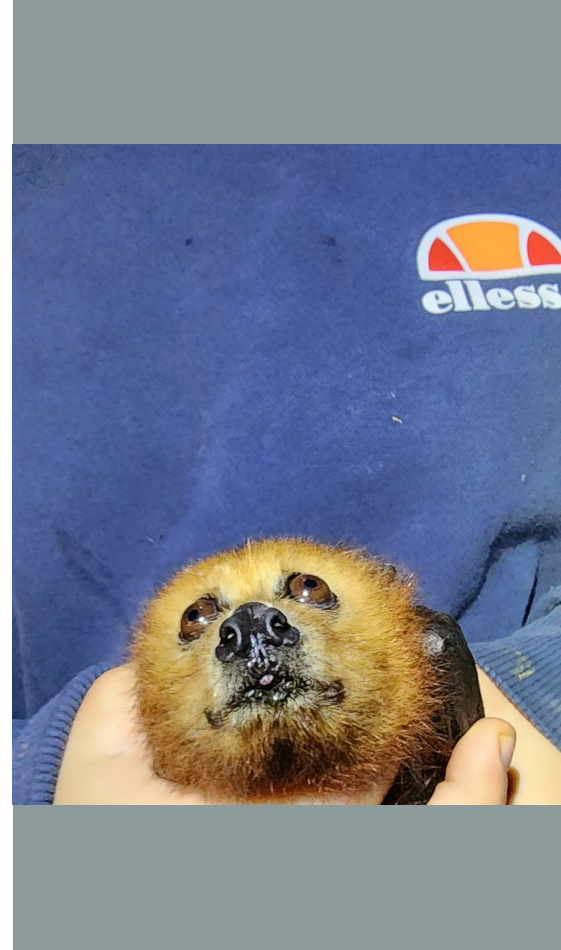
Multi-Angle + Augmentation (Same Frame)



Modified color



Face cut



Repositioning



Rotated

Frontal Only (Different bats)



bat_1



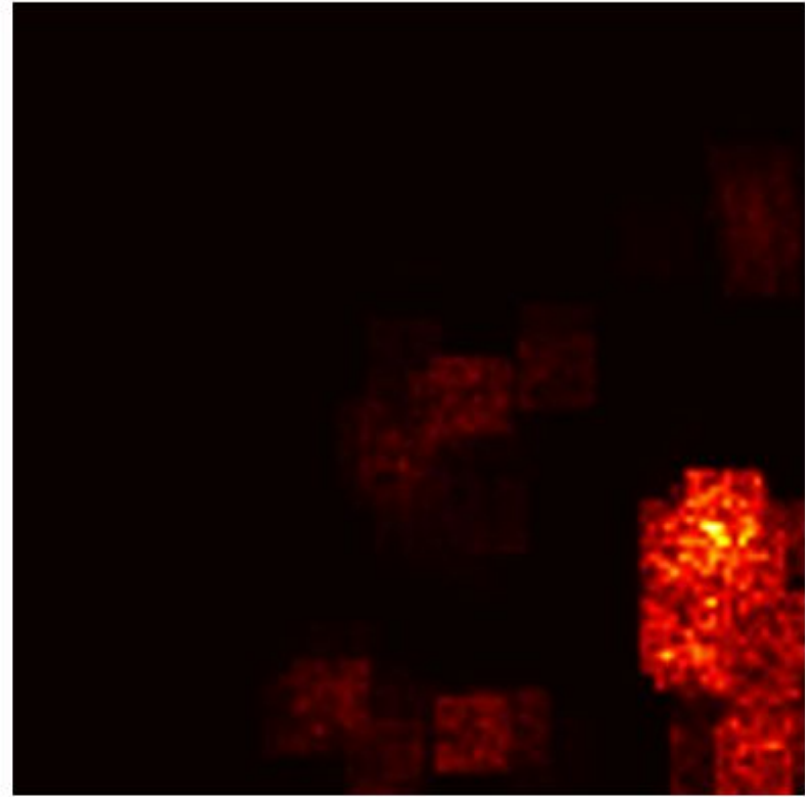
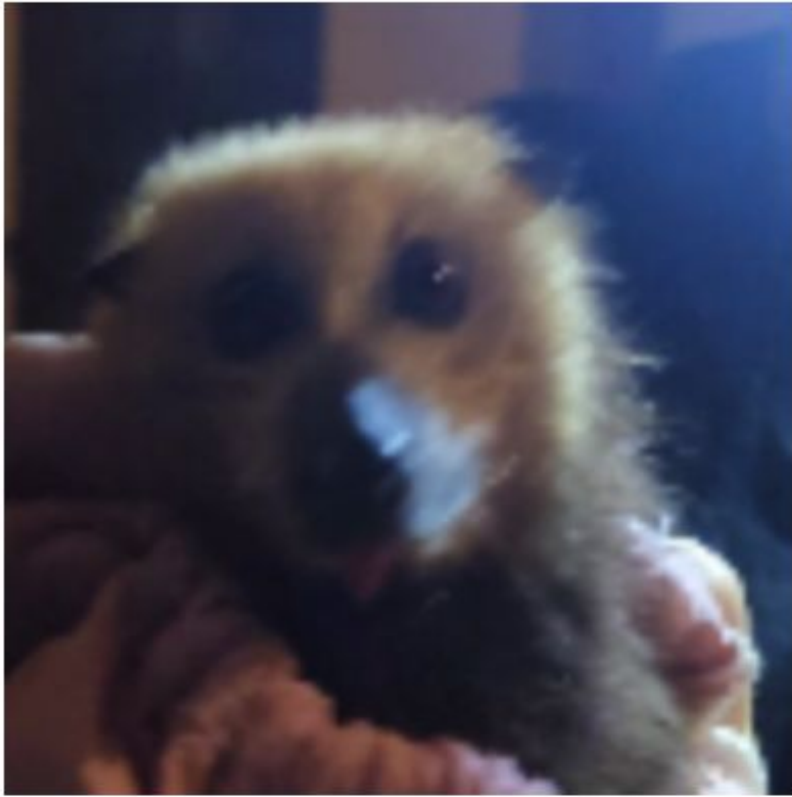
bat_2



bat_3

Frontal Only (Different bats)

20230807_060504.1413.png



Frontal Only (Green Background)



bat_1



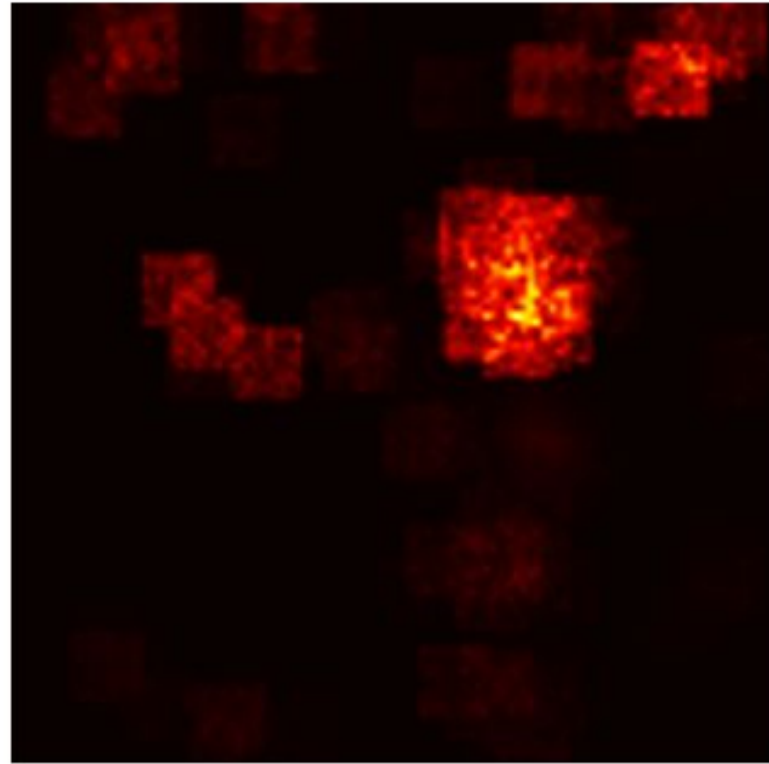
bat_2



bat_3

Frontal Only (Green Background)

20230805_145544.1064.jpg



Frontal Only (Random Background)



bat_1



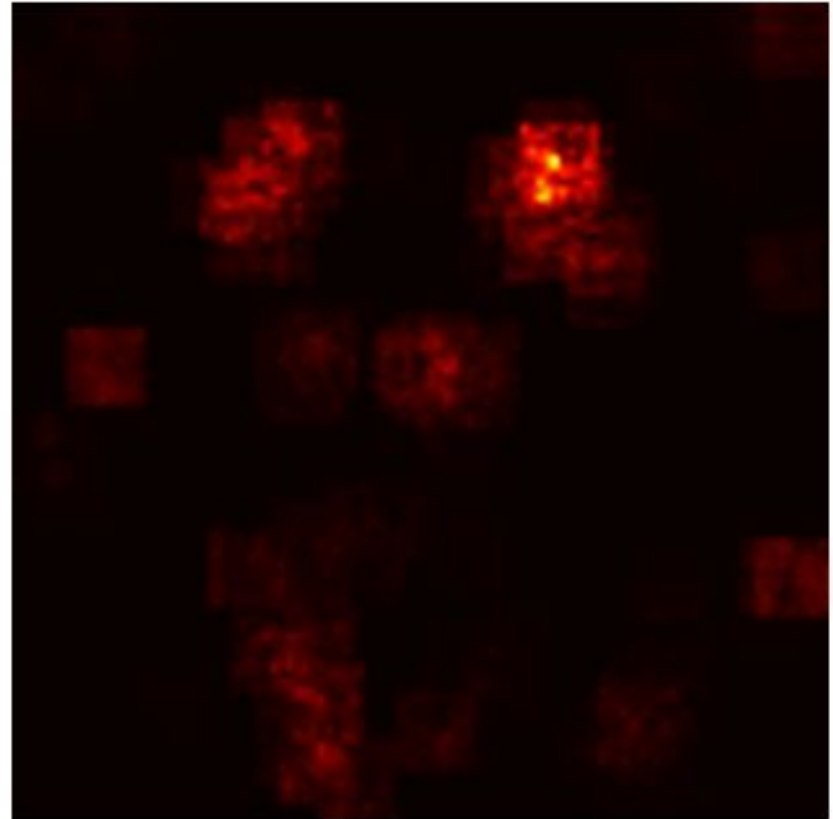
bat_2



bat_3

Frontal Only (Random Background)

831_141252/20230831_141252.292.png



bat_1

Frontal Images

Experiments Findings

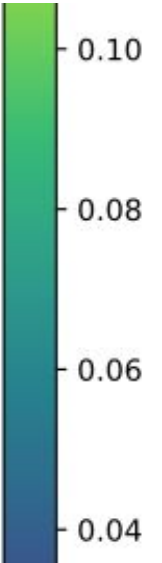
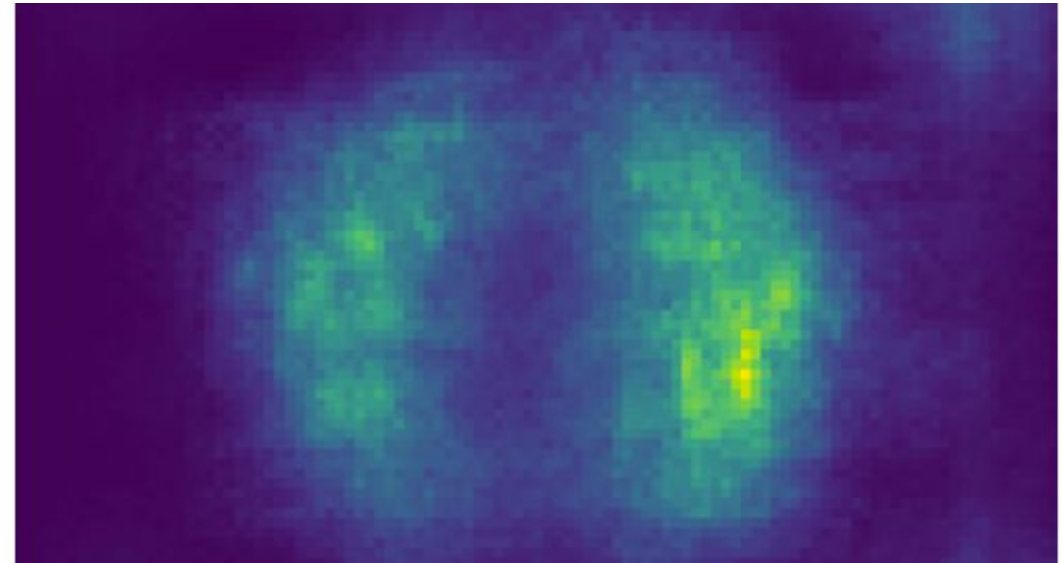
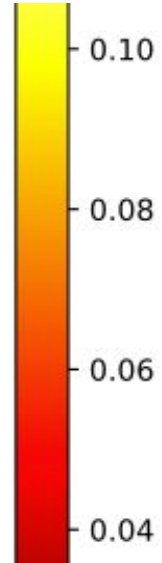
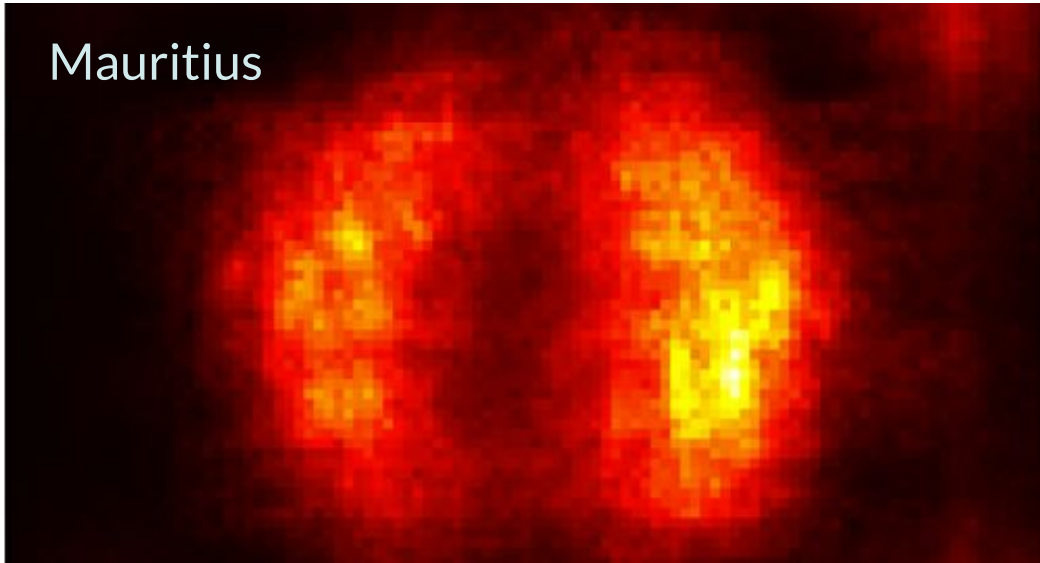
Although green background images had the highest prediction scores per bat, the model didn't generalize well. Random backgrounds gave slightly lower scores but generalized better to unseen bats.



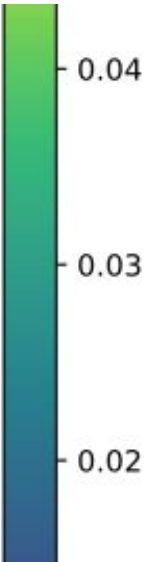
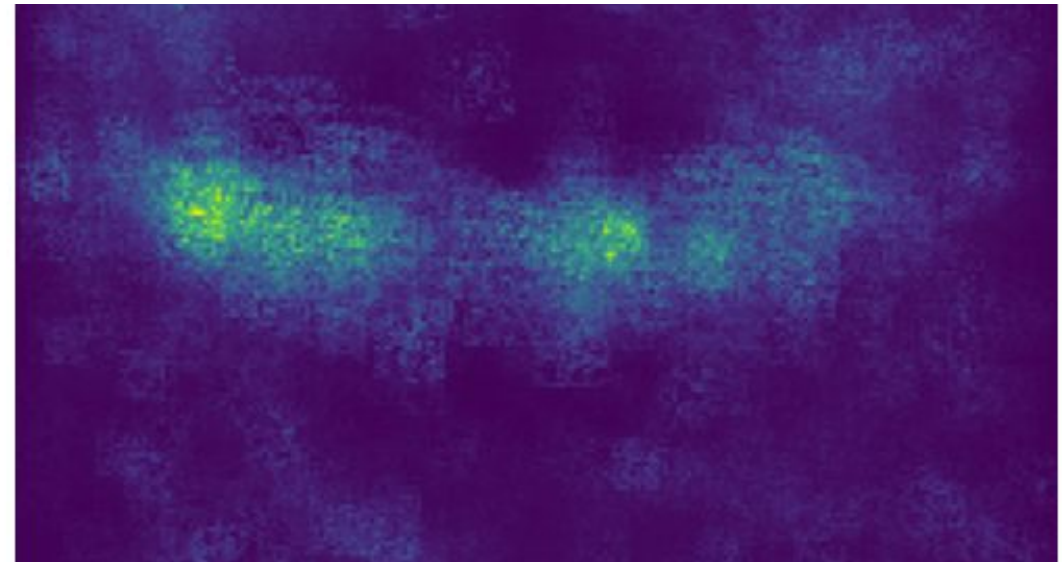
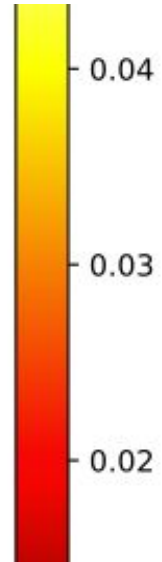
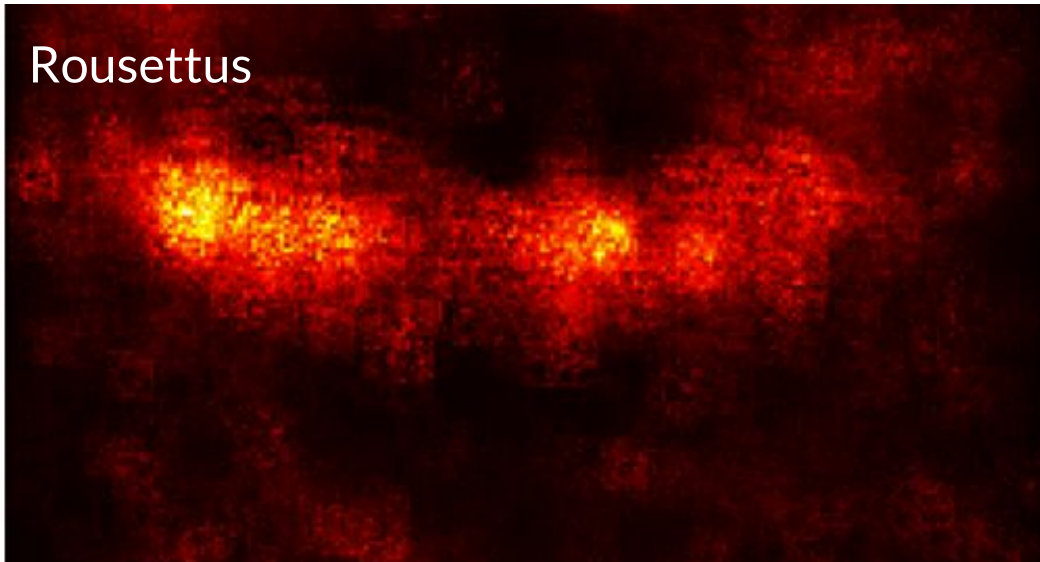
[illegible]

Mean Saliency Maps

Mauritius



Rousettus



What is MLFlow?



Introduction to MLflow



mlflow

MLflow is an open-source platform to manage the machine learning lifecycle.

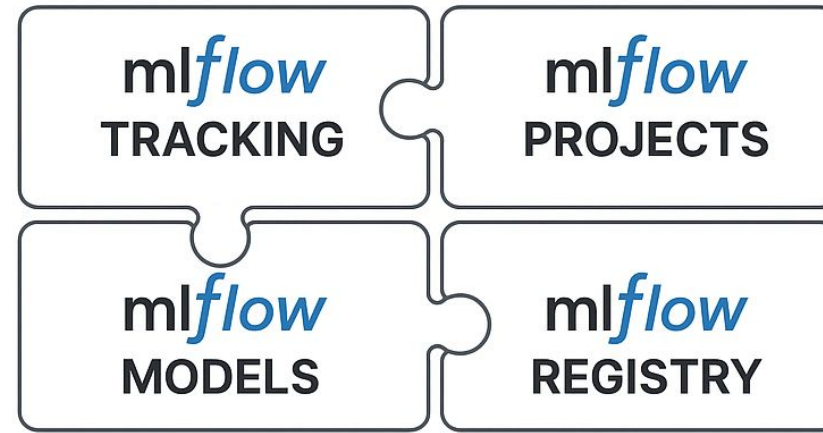
- Tracking experiments
- Packaging code
- Sharing models
- Deploying to production

Why use MLFlow?

Use MLFlow?	NO!	Yes :))
Parameters logging	Parameters lost in notebooks	Parameters auto-logged
Model versions tracking	Confusing model versions	Model versions tracked
Reproducibility	Hard to reproduce results	Full reproducibility
Results Comparison	Manual result comparison	Interactive UI comparison

MLFlow components

MLflow Components



1. **MLflow Tracking** – Log parameters, metrics, and artifacts.
2. **MLflow Projects** – Package code in a reproducible format.
3. **MLflow Models** – Manage and deploy trained models
4. **MLflow Registry** – Centralized model store with versioning

Getting started

Install

```
pip install mlflow
```

Log Your First Experiment

```
import mlflow

mlflow.log_param("lr", 0.01)

mlflow.log_metric("accuracy", 0.9)
```

Examine the results

```
mlflow ui
```

Conclusion

